

CNSCC.160 – Fundamentals of Communication Systems

Tutorial Questions and Revision (Secure Communication)

Knowledge

Q1. Which of the following words mean ‘secret writing’? Which one means ‘covered writing’?

- a. cryptography
- b. steganography

Q2. Alice has found a way to write secretly to Bob. Each time, she takes a new text, such as an article from the newspaper, but inserts one or two spaces between the words. A single space means a binary digit 0; a double space means a binary digit 1. Bob extracts the binary digits and interprets them using ASCII code. Is this an example of cryptography or steganography? Explain.

Q3. Alice and Bob exchange confidential messages. They share a very large number as the encryption and decryption key in both directions. Is this an example of symmetric-key or asymmetric-key cryptography? Explain.

Q4. Distinguish between a substitution cipher and a transposition cipher.

Q5. In a cipher, all the As in the plaintext have been changed to Ds in the ciphertext and all the Ds in the plaintext have been changed to Hs in the ciphertext. Is this a monoalphabetic or polyalphabetic substitution cipher? Explain.

Q6. Which cipher can be broken more easily, monoalphabetic or polyalphabetic?

Q7. Assume Alice and Bob use an additive cipher in modulo 26 arithmetic. If Eve, the intruder, wants to break the code by trying all possible keys (brute-force attack), how many keys should she try on average?

Q8. Assume we have a plaintext of 1000 characters. How many keys do we need to encrypt or decrypt the message in each of the following ciphers?

- a. additive
- b. monoalphabetic
- c. autokey

Q9. Why do you think asymmetric-key cryptography is used only with small messages?

Q10. In an asymmetric public key cipher, which key is used for encryption? Which key is used for decryption?

- a. public key
- b. private key

Q11. What is the role of the secret key added to the hash function in Figure (on slide 11, week 8, part2) in the text (MAC)? Explain.

Q12. Distinguish between message authentication and entity authentication.

Q13. Alice signs the message she sends to Bob to prove that she is the sender of the message. Which of the following keys does Alice need to use?

- a. Alice's public key
- b. Alice's private key

Q14. Which of the following services are not provided by digital signature?

- a. message authentication
- b. confidentiality
- c. nonrepudiation

Understanding

Q1. Define the type of attack in each of the following cases:

- a. A student breaks into a professor's office to obtain a copy of the next test.
- b. A student gives a check for \$10 to buy a used book. Later the student finds out that the check was cashed for \$100.
- c. A student sends hundreds of emails per day to the school using a phony return email address.

Q2. Use the additive cipher with $k = 10$ to encrypt the plaintext 'book'. Then decrypt the message to get the original plaintext.

Q3. Encrypt the message 'this is an exercise' using additive cipher with key = 20. Ignore the space between words. Decrypt the message to get the original plaintext.

Q4. *Atbash* was a popular cipher among Biblical writers. In *Atbash*, 'A' is encrypted as 'Z', 'B' is encrypted as 'Y', and so on. Similarly, 'Z' is encrypted as 'A', 'Y' is encrypted as 'B', and so on. Suppose that the alphabet is divided into halves and the letters in the first half are encrypted as the letters in the second and *vice versa*. Find the type of cipher and key. Encipher the plaintext 'an exercise' using the *Atbash* cipher.

Q5. Alice can use only the additive cipher on her computer to send a message to a friend. She thinks that the message is more secure if she encrypts the message two times, each time with a different key. Is she right? Defend your answer.

Q6. One of the attacks an intruder can apply to a simple cipher like an additive cipher is called the *ciphertext* attack. In this type of attack, the intruder intercepts the cipher and tries to find the key and eventually the plaintext. One of the methods used in a ciphertext attack is called the *brute-force* approach, in which the intruder tries several keys and decrypts the message until the message makes sense. Assume the intruder has intercepted the ciphertext 'UVACLYZLJBYL'. Try to decrypt the message by using keys from 1 until a plaintext appears that makes sense.

Q7.

- (i) What is the maximum number of keys that an eavesdropper needs to try in order to unscramble a ciphertext which has been encrypted using the Caesar Cipher?
- (ii) What would your answer be if a randomly-ordered cipher alphabet was employed?
- (iii) How could we try and crack a cipher based on a randomly-ordered cipher alphabet?

Q8. A small private club has only 100 members:

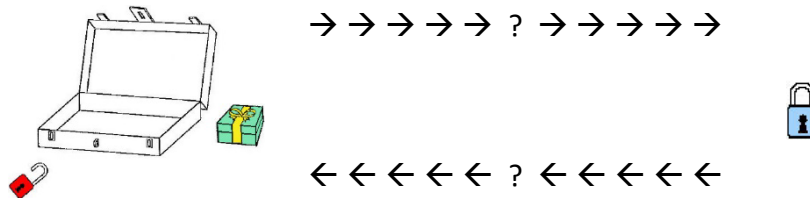
How many secret keys are needed if all members of the club need to send secret messages to each other?

How many secret keys are needed if everyone trusts the president of the club? If a member needs to send a message to another member, she first sends it to the president; the president then sends the message to the other member.

Q9. Symmetric-key cryptography is analogous to the exchange of a briefcase that contains a gift. The sender uses a padlock to lock the briefcase and hires a courier to deliver it to the intended receiver. A copy of the key is also sent to the receiver via a secure channel. Consider the case of asymmetric-key cryptography, where the sender owns a (red) padlock and a key, whilst the receiver owns a different (blue) padlock and a key. How should the sender have the briefcase delivered to the receiver without the requirement of key exchange?

Sender

Receiver



Q10. Use the Playfair cipher to encipher the message "The key is hidden under the door pad". The secret key can be made by filling the first and part of the second row with the word "GUIDANCE" and filling the rest of the matrix with the rest of the alphabet.

Hint:

G	U	I/J	D	A
N	C	E	B	F
H	K	L	M	O
P	Q	R	S	T
V	W	X	Y	Z

Please note that

1. If the two letters in the pair are located in the same row of the secret key, the corresponding encrypted character for each letter is the next letter to the right in the same row (in case of last char- wrapping to the beginning of the row)
2. If the two letters in the pair are located in the same column of the secret key, the corresponding encrypted character for each letter is the letter beneath it, in the same column (with wrapping if the last one).
3. If the two letters in the pair are not in the same row or column of the secret key, the corresponding encrypted character for each letter is a letter that is in its own row but in the same column as the other letter.